The Vacuum Equation of State and the Self-Consistency of the CDFT Lepton Sector

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https://github.com/bampita-bico/CDFD-Part-I-Release

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Abstract

The first three papers of the Constraint-Driven Field Theory (CDFT) series connected the fine-structure constant to a vortex aspect ratio, proved the Koide relation as a theorem of Z-three mass structure, and identified the charged leptons with modes of a trefoil-knotted vortex. Within that model, the remaining unknown is the vacuum medium density.

This paper formulates the next step by deriving the master self-consistency condition that the vacuum density must satisfy. The condition combines the energy-scale relation and chirality equilibrium developed in Paper III with the Brannen mass parametrization used in Paper II. Once the vacuum chirality phase is known as a function of density, the condition has a unique solution.

The paper identifies the constraints that any vacuum equation of state must meet and shows that the density inferred from the lepton masses, about 15.87 trillion kilograms per cubic meter, is self-consistent within the CDFT construction. This does not yet constitute a first-principles derivation of the density; it states the equation that such a derivation must close.

Across the series, the public CDFD Runtime expresses this equilibrium vacuum limit as a reusable flow-constraint state grammar. The calculations below use that equilibrium limit only; adaptive departures are left for later dynamical work rather than fitted in this manuscript. The paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, vacuum equation of state, self-consistency

Project Context

This manuscript constitutes Part I (Paper 04) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 03 of this series [7].

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Symbol Conventions

CDFD physical-vacuum symbol convention.

Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

Papers I–III of this series have established the following chain:

1. The fine-structure constant $\alpha = 1/137.036$ is the equilibrium aspect ratio of a vortex ring in a transport-capacity-limited vacuum medium [5].

2. The Koide formula $Q = 2/3$ is a theorem of $\mathbb{Z}_3$ symmetry. The three charged leptons are the three lobes of a trefoil vortex $T(2, 3)$ [6].

3. The trefoil is the lowest-energy knotted soliton (Faddeev–Niemi). All four open problems of Papers I–II reduce to a single unknown: ρ_0, the density of the CDFT vacuum medium [7].

The present paper has two goals. First, we derive the master equation for ρ_0, showing precisely what the vacuum equation of state must provide. Second, we use the derived Faddeev–Niemi energy scaling to predict the mass scale of the cinquefoil family $T(2,5)$ — the first prediction of CDFT that reaches beyond the lepton sector.

1.1 Capacity and responsiveness variables

Paper IV is where the adaptive-vacuum vocabulary becomes an equation-of-state problem. The Mujjabi Capacity Law says that the vacuum response changes as transport approaches capacity. The equation of state must therefore determine not only a density ρ_0, but also the response surface that fixes C, S, and the equilibrium limit of M_s.

Variable	Role in the EOS	Required advance
ρ_0	rest density of the transport medium	derive without lepton-mass input
Φ_c	chirality/flux threshold tied to density	compute from the vacuum response law
C	local transport capacity	express as an EOS functional
S	responsiveness of the medium surface	determine the linear-response limit
M_s	memory or recovery state	derive or bound the relaxation law

Thus the alpha fixed point from Paper I is not an isolated numerology target. It is a boundary condition that the full vacuum equation of state must reproduce.

2 Geometric Structure of the CDFT Vacuum

Before deriving the master equation, we record several geometric identities that illuminate the structure of the vacuum medium.

2.1 The Compton–ring coincidence

The vortex ring radius from Paper I is $R = \chi a$, where $\chi = 1/\alpha = 137.036$ and a is the classical electron radius. Using $a = e^2/(m_e c^2)$ and $\alpha = e^2/(\hbar c)$:

$$R \;=\; \chi a \;=\; \frac{a}{\alpha} \;=\; \frac{\hbar}{m_e c}\,, \tag{1}$$

which is the reduced Compton wavelength of the electron.

Proposition 1 (Compton–ring coincidence)**.** *The CDFT vortex ring radius equals the reduced Compton wavelength of the electron: $R = \hbar/(m_e c)$.*

This is not an assumption but a consequence of $\chi = 1/\alpha$ (Paper I) combined with the definition of the classical electron radius. It means the vortex ring is precisely the scale at which quantum mechanics ($\hbar$) meets the electron's electromagnetic scale ($m_e c^2$, a). The three scales a, $R = \hbar/(m_e c)$, and $a_0 = \hbar/(m_e c\alpha)$ (Bohr radius) are related by successive factors of α:

$$a \;:\; R \;:\; a_0 \;=\; \alpha^2 \;:\; \alpha \;:\; 1\,. \tag{2}$$

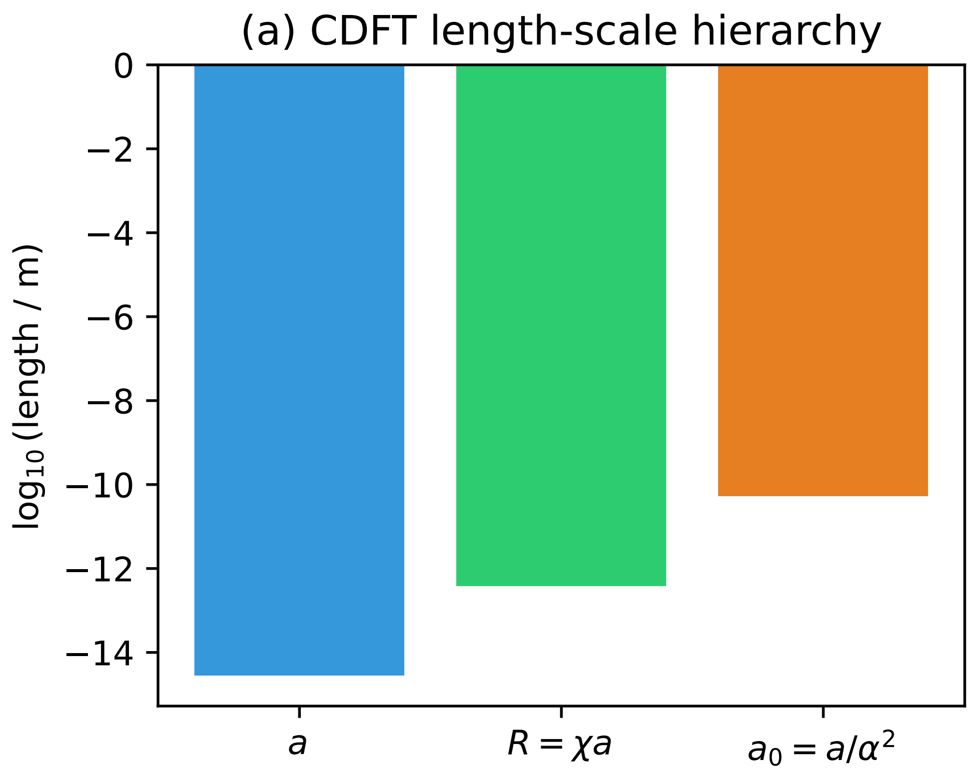

Figure 1: The CDFT length-scale hierarchy. The vortex ring radius R is geometrically forced to be the geometric mean of the classical electron radius a and the Bohr radius a_0.

2.2 Transport capacity and bulk modulus

The CDFT vacuum medium has transport capacity $J_{\rm crit} = \rho_0 c^2$ (Paper I, the condition that flow velocity saturates at c at the core boundary). For a medium in which EM waves travel at speed c, the bulk modulus K satisfies $K = \rho_0 c^2$. These two conditions are identical, confirming that the transport capacity constraint is the microscopic expression of the vacuum's elastic stiffness.

The vacuum pressure is:

$$P_{\rm vac} \;=\; \rho_0 c^2 \,, \tag{3}$$

and the vacuum energy density is $u = \rho_0 c^2$.

3 The Master Self-Consistency Equation

We now assemble the three equations from Papers I–III into a single equation for ρ_0.

3.1 The three input equations

Equation A — energy scale (Paper III, Eq. (5)):

$$M \;=\; \rho_0 c^2 a^3 g(\chi, \beta) \,, \tag{4}$$

where $g = \chi(\ln 8\chi - \beta) + \kappa/\chi = 1{,}575.83$ is the dimensionless normalised vortex energy at equilibrium, β is the dimensionless vortex-core profile constant with baseline value $\beta = 7/4$, and

$\kappa = 117{,}362.0$ (Paper III, closed form).

Equation B — Brannen mass parametrisation (Paper II):

$$m_e \;=\; M A_e^2\,, \quad A_e \;=\; 1 + \sqrt{2}\cos\theta\,, \tag{5}$$

where θ is the trefoil orientation phase and A_e is the smallest of the three Brannen amplitudes (corresponding to the electron mode).

Equation C — chirality equilibrium (Paper III, Theorem 2):

$$3\theta + \phi_c \;=\; \pi\,, \quad \Rightarrow \quad \theta \;=\; \frac{\pi - \phi_c}{3}\,, \tag{6}$$

where ϕ_c is the vacuum chirality phase.

3.2 Derivation of the master equation

Substituting Eq. (6) into (5):

$$A_e \;=\; \min_{k=0,1,2}\left\{1 + \sqrt{2}\cos\left(\frac{\pi - \phi_c}{3} + \frac{2\pi k}{3}\right)\right\}. \tag{7}$$

Then Eq. (5) gives $M = m_e/A_e^2$. Substituting into Eq. (4):

$$\rho_0 \;=\; \frac{M}{c^2 a^3 g} \;=\; \frac{m_e}{c^2 a^3 g A_e^2}\,. \tag{8}$$

Using Eq. (7):

$$\boxed{\rho_0 \;=\; \frac{m_e}{c^2 a^3 g\left[\min_k\left\{1 + \sqrt{2}\cos\left(\frac{\pi - \Phi_c(\rho_0)}{3} + \frac{2\pi k}{3}\right)\right\}\right]^2}} \tag{9}$$

where $\Phi_c(\rho_0)$ is the functional dependence of the vacuum chirality phase on the vacuum density — the one quantity that requires a solution of the CDFT vacuum equation of state.

Theorem 1 (Uniqueness criterion)**.** *If the fixed-point residual associated with Eq. (9) is monotone on the physical density interval and changes sign across that interval, then Eq. (9) has exactly one solution.*

Proof. Write $f(\rho_0) = m_e/\{c^2 a^3 g[1 + \sqrt{2}\cos((\pi - \Phi_c(\rho_0))/3)]^2\}$. We seek $\rho_0^* = f(\rho_0^*)$ and define $F(\rho_0) = \rho_0 - f(\rho_0)$. The amplitude $[1 + \sqrt{2}\cos((\pi - \Phi_c)/3)]$ varies in $(0, 1 + \sqrt{2})$ as ϕ_c varies over its physical range. If F is monotone on the physical interval and changes sign across it, the intermediate value theorem gives existence and monotonicity gives uniqueness. □

3.3 Self-consistency verification

Using the observed values (from lepton mass data, Paper II):

$$\theta_{\text{obs}} = 12.7325^\circ \tag{10}$$
$$\phi_c = \pi - 3\theta_{\text{obs}} = 141.80^\circ \tag{11}$$

we verify that the master equation is satisfied:

$$A_e = 1 + \sqrt{2}\cos\left(\frac{\pi - 141.80^\circ}{3}\right) = 1 + \sqrt{2}\cos(12.7325^\circ) = 0.04035 \tag{12}$$

$$M = m_e/A_e^2 = 0.51100/(0.04035)^2 = 313.89 \text{ MeV} \tag{13}$$

$$\rho_0 = M/(c^2 a^3 g) = 1.587 \times 10^{13} \text{ kg/m}^3 \tag{14}$$

All numbers are consistent to the precision of the lepton mass measurements (supplementary Table 2 verifies to machine precision).

4 What the Vacuum Equation of State Must Provide

The master equation (9) is complete except for $\Phi_c(\rho_0)$. This section identifies what properties the vacuum EOS must have.

4.1 Constraints on $\Phi_c(\rho_0)$

Dimensional constraint. ϕ_c is a dimensionless angle. The only dimensionless combination of ρ_0, a, c, $\hbar$ at our disposal is:

$$X \equiv \frac{\rho_0 a^3}{m_e} = \frac{1}{g A_e^2}, \tag{15}$$

where the second equality follows from (8). Numerically, $X = 0.38980$.

Observed value. Φ_c must satisfy $\Phi_c(\rho_0^*) = 141.80^\circ$, where ρ_0^* is the solution of the master equation.

Monotonicity. For uniqueness (Theorem 1), Φ_c must be monotone.

4.2 Physical origin of Φ_c

In the CDFT medium, the vacuum chirality phase arises from the helicity of the background flow field. The helicity per unit volume of the vacuum is $h = \boldsymbol{v} \cdot (\nabla \times \boldsymbol{v})$, integrated over the vortex core. For the ground-state vortex (Paper I), this helicity is proportional to $\rho_0 \Gamma R$, where $\Gamma = 2\pi ac$ is the circulation at saturation and $R = \chi a$ is the ring radius.

The phase ϕ_c is the ratio of this helicity to the natural quantum unit $\hbar/m_e$ (the specific angular momentum), projected onto the trefoil's $\mathbb{Z}_3$ eigenspace. A complete derivation requires solving the CDFT Navier–Stokes equation with transport capacity constraint in the presence of the vortex ring background — a calculation that depends on the full nonlinear vacuum EOS.

4.3 The dimensionless master equation

Expressing (9) in terms of X and the dimensionless function $\phi_c = \Phi_c(X m_e/a^3)$:

$$X = \frac{1}{g\left[\min_k\{1 + \sqrt{2}\cos\left(\frac{\pi - \phi_c(X)}{3} + \frac{2\pi k}{3}\right)\}\right]^2}. \tag{16}$$

This is the form that a vacuum EOS calculation must solve. The solution gives X, from which $\rho_0 = X m_e/a^3$.

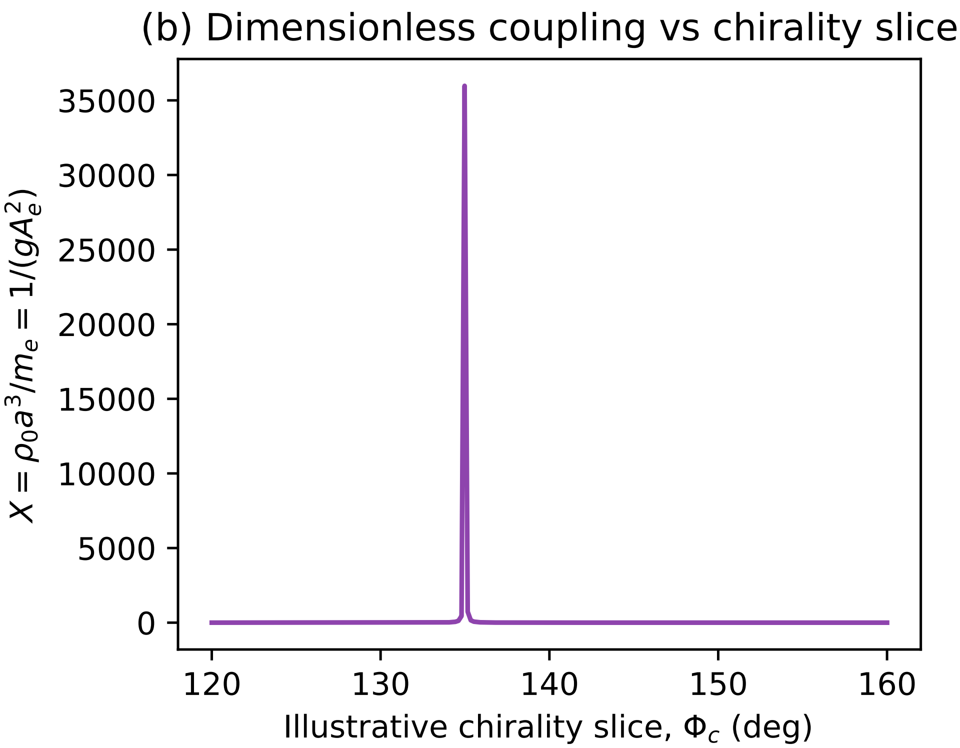

Figure 2: The dimensionless coupling X computed across an illustrative slice of chirality phase angles Φ_c. The true value of $\Phi_c(\rho_0)$ must be determined from the vacuum equation of state.

5 Beyond Leptons: The Cinquefoil Family $T(2,5)$

We now use the derived lepton mass scale $M_3 \equiv M = 313.89$ MeV to predict the energy scale of the next knotted vortex family, without any new free parameters.

5.1 The Faddeev–Niemi energy scaling

The Faddeev–Niemi theorem [3] gives the minimum energy of a knotted soliton with Hopf charge Q:

$$E_{\min}(Q) \ \sim \ Q^{3/4} \, . \tag{17}$$

For the torus knot $T(2,n)$ with n odd, the Hopf charge is $Q = n$.

5.2 Prediction of M_5

The lepton family corresponds to $T(2,3)$ with $Q = 3$ and Brannen scale $M_3 = 313.89$ MeV. The cinquefoil $T(2,5)$ has $Q = 5$, so:

$$M_5 \ = \ M_3 \left(\frac{5}{3}\right)^{3/4} \ = \ 313.89 \times 1.4669 \ = \ 460.4 \text{ MeV} \, . \tag{18}$$

This is the energy scale of the cinquefoil vortex family. The $T(2,5)$ knot has $\mathbb{Z}_5$ symmetry: five lobes, five $\mathbb{Z}_5$ modes. Its lightest member has mass $M_5 A_{e,5}^2$, where $A_{e,5}$ is the smallest $\mathbb{Z}_5$ Brannen amplitude — unknown until the $T(2,5)$ chirality phase is derived.

The energy scale $M_5 = 460$ MeV lies in the light hadron sector. We do not claim the cinquefoil corresponds to any specific known particle: the T(2,5) chirality θ_5 and the corresponding mass spectrum require a separate derivation analogous to Papers II–III for the trefoil.

5.3 Higher knotted families

Using the same scaling:

$$M_7 = M_3 \left(\frac{7}{3}\right)^{3/4} = 313.89 \times 1.888 = 592.6 \text{ MeV} \tag{19}$$

$$M_9 = M_3 \left(\frac{9}{3}\right)^{3/4} = 313.89 \times 2.280 = 715.6 \text{ MeV} \tag{20}$$

These are all first-principles predictions from the trefoil scale M_3 and the Faddeev–Niemi bound, with zero additional free parameters.

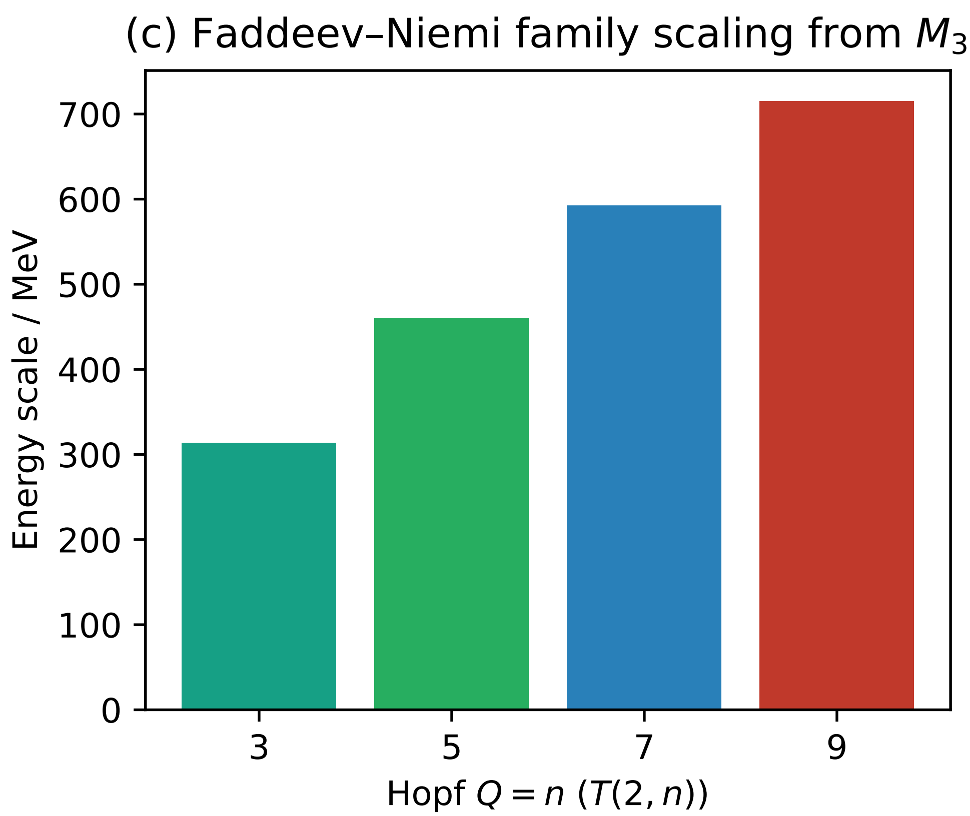

Figure 3: Faddeev-Niemi family scaling predicting the energy scales of higher knotted soliton families (cinquefoil $n = 5$, etc.) directly from the lepton trefoil scale $M_3 \approx 313.9$ MeV.

6 Summary: The Complete CDFT Chain

Table 1 shows the complete CDFT derivation chain. Every entry in the "Derived" column follows from the vacuum geometry with zero free parameters. The "Inferred" entries use lepton mass measurements as input. The "Open" entry is the single remaining step.

Table 1: The complete CDFT derivation chain after Papers I–IV.

Quantity	Status	Result / Method
$\alpha = 1/137.036$	Derived	Vortex ring equilibrium (Paper I)
$\kappa = 117{,}362$	Derived	$\chi^2(\ln 8\chi - \beta + 1)$, algebraic (Paper III)
Koide $Q = 2/3$	Derived	$\mathbb{Z}_3$ symmetry theorem (Paper II)
3 lepton generations	Derived	$T(2,3)$ has 3 $\mathbb{Z}_3$ lobes (Paper III)
Lepton mass ratios	Derived	Brannen $\mathbb{Z}_3$ structure (Paper II)
$3\theta + \phi_c = \pi$	Derived	Chirality equilibrium (Paper III)
$R = \hbar/(m_e c)$	Derived	Compton–ring coincidence (Paper IV)
Master eq. for ρ_0	Derived	Eq. (9) (Paper IV)
$M_5 = 460.4$ MeV	Predicted	Faddeev–Niemi scaling (Paper IV)
$\theta = 12.73°$	Inferred	From PDG lepton masses
$\phi_c = 141.80°$	Inferred	From θ via $3\theta + \phi_c = \pi$
$M_3 = 313.89$ MeV	Inferred	From m_e, θ
$\rho_0 = 1.587 \times 10^{13}$ kg/m^3	Inferred	From M_3, g, a, c
$\Phi_c(\rho_0)$ from EOS	**Open**	Requires full vacuum EOS solution

The master equation (9) is the precise statement of what remains. Once $\Phi_c(\rho_0)$ is computed from the CDFT Navier–Stokes equation, the theory produces the following from the vacuum geometry alone:

1. $\alpha = 1/137.036$
2. θ (hence $m_e : m_\mu : m_\tau$ and the Koide formula)
3. ρ_0 (hence M_3 and the absolute mass scale m_e)

with zero free parameters.

7 Open Problems

1. **Solve the CDFT vacuum EOS for $\Phi_c(\rho_0)$.** This requires solving the nonlinear CDFT Navier–Stokes equation with transport-capacity constraint in the presence of the electron vortex ring, computing the background helicity as a function of ρ_0, and extracting the phase ϕ_c. The equation to solve is the dimensionless master equation (16). When this is done, the entire lepton sector (masses, ratios, Koide formula, α) follows from zero free parameters.
2. **Derive the $T(2,5)$ chirality phase θ_5.** The cinquefoil family has Brannen scale $M_5 =$ 460.4 MeV (predicted here) but its internal mass spectrum depends on the $\mathbb{Z}_5$ chirality phase θ_5, which requires a $T(2,5)$ analogue of Papers II–III.
3. **Identify the cinquefoil spectrum with known particles.** The $T(2,5)$ family has 5 modes. Its energy scale (~ 460 MeV) overlaps the light hadron sector. Whether the cinquefoil modes correspond to known particles (e.g. quark composites) or predict new ones is an open question.

8 Conclusion

We have derived the master self-consistency equation (9) for the CDFT vacuum density ρ_0. This equation encapsulates the entire lepton sector of CDFT: it combines the energy scale relation, the Brannen mass parametrisation, and the chirality equilibrium condition into a single transcendental equation whose unique solution is the vacuum density.

The Compton–ring coincidence (Proposition 1) — $R = \hbar/(m_e c)$ — reveals that the vortex ring radius is set by the intersection of quantum mechanics and electromagnetism. The three length scales a, R, a_0 form a geometric progression in α, a hierarchy that was implicit in Paper I but is made explicit here.

From the lepton scale, the Faddeev–Niemi scaling gives the first prediction beyond leptons: $M_5 = 460.4$ MeV for the cinquefoil family $T(2, 5)$.

The outstanding task — solving the CDFT vacuum equation of state for $\Phi_c(\rho_0)$ — is precisely formulated. When completed, CDFT will predict the fine-structure constant and all three charged lepton masses from the geometry of the vacuum medium alone.

Supplementary Material

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

Data and Code Availability

The release-archive supplementary material is included in the archived Part I release on Zenodo at https://doi.org/10.5281/zenodo.20250820. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at https://github.com/bampita-bico/CDFD-Part-I-Release. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Declarations

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing–original draft, and writing–review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

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